VINITHA M RAJAN

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EXECUTIVE SUMMARY

I am a PhD scholar in the Department of Data Science, with research interests in time-series analysis, anomaly detection, deep learning, and Explainable AI. My research focuses on developing reliable and interpretable learning methods for complex multivariate time-series data with applications in industrial and real-world systems. In particular, anomaly detection is an important requirement for the preventive maintenance of many real-world critical systems. My broader research goal is to develop robust and explainable AI methods that combine technological advances in machine learning with practical solutions to challenging, real-world problems. I am also open to exploring other frontiers and modalities in data science.

AREAS OF INTEREST

Time Series Analysis; Anomaly Detection; Data Analytics; Ethical AI; Text Mining

ACADEMIC PROFILE

PhD Scholar, Department of Data Science **2021 - Now**
Indian Institute Of Technology Palakkad.
CGPA : 9.75/10

Master of Technology in Computer Science & Engineering **2017 - 2019**
College Of Engineering Trivandrum.
CGPA : 9.47/10

Bachelor of Technology in Computer Science & Engineering **2004 - 2008**
National Institute Of Technology Calicut.
CGPA : 8.48/10

PUBLICATIONS

Vinitha M Rajan, Sahely Bhadra, **HEIGHTS: Hierarchical graph structure learning for time series anomaly detection**, Journal Of Neurocomputing, Volume 671, 2026, 132638, ISSN 0925-2312, `https://doi.org/10.1016/j.neucom.2026.132638`.

Rajan V.M., Ramanujan A., (2021) **Architecture of a Semantic WordCloud Visualization**. Second International Conference on Networks and Advances in Computational Technologies. Transactions on Computational Science and Computational Intelligence. Springer, DOI: 10.1007/978-3-030-49500-8_9.

RESEARCH PROJECTS

1. **Multi-Variate Time Series Anomaly Detection** **Aug'21 - Now**
   I have explored three research directions to develop unsupervised methods for anomaly detection in multivariate time-series data from cyber-physical systems as follows

   - **Hierarchical Graph Structure Learning:** In our first published work, named HEIGHTS (Rajan & Bhadra, 2026), we developed a first of its kind to model the subsystem architecture and their interactions in the underlying systems. We applied the principle of multi-view learning by constructing a two-level hierarchical structure. The two-level hierarchy represents variables/features at the base level and feature groups at the higher level. We conducted extensive experiments on both synthetic and real datasets to validate our hypotheses on hierarchical structure learning and anomaly detection performance. Our methodology has been proven to be an efficient and robust approach for diagnosing potential hazards in realistic industrial systems.

- **Nonparametric Feature Clustering:** While in the first work we learned and utilized relationships among groups of variables (features), with the groups predefined in their respective systems, we formulated the second objective to leverage model-informed decisions about feature groups. In this direction, we developed an unsupervised deep learning mechanism named FILTER that automatically clusters (groups) the features and uses this information for anomaly detection. We demonstrated that FILTER achieves competitive performance and serves as an explainable method by evaluating its effectiveness in capturing root-cause features during an anomalous instant. The work is currently under review.
- **Explainable Anomaly Detection:** While the explainability in the second work focused on identifying the root cause features in the event of an anomaly, it would be more advantageous if the end users (researchers or operations engineers) can provide an expected pattern that the anomalous sample has violated. Hence, in this research direction, I take a further step toward explainability by identifying a set of prototypes that the data exhibits. These prototypes are representative of the system's normal behavior; hence, with this additional information, we aim to improve detection performance and interpretability for the end users.

2. **Fast Radio Burst (FRB) Identification In Astronomical Time Series Data** **Nov'22 - Jan'24**
This research project, conducted in collaboration with NCRA Pune, NVIDIA, and IIT Palakkad under the NVIDIA AI Technology Center (NVAITC) program, focused on developing a multi-beam AI classifier to detect Fast Radio Bursts (FRBs). The work involved preprocessing large volumes of telescopic time-series data and extracting key parameters (width, dispersion measure, Signal-To-Noise Ratio, and arrival time). Nonparametric clustering was applied to the dispersion measure and time features to identify candidate FRBs, followed by linear regression to find the cluster that best matches the expected DM-time relationship based on the underlying physics laws. The approach yielded satisfactory results, advancing AI techniques for large-scale data processing and real-time inference on high-throughput streams. The work directly contributed to the SPOTLIGHT system deployed at the GMRT, NCRA, Pune.

3. **TCAV (Testing With Concept Activation Vector)** **April'22**
As part of the "CS6006: Responsible AI" coursework at IIT Palakkad, this project applied TCAV—an interpretability technique developed by Google—to identify human-friendly concepts (shape, colour, texture) most relevant to a given fruit object. A fruit classifier was built by transfer-learning a pretrained GoogleNet model on the Fruits360 dataset. Human-understandable concepts were defined using representative image sets, and concept activation vectors (directions) were learned for each. TCAV scores were then computed to measure how well each concept was captured within the network's hidden layers. The results confirmed that the model successfully learned and associated the intended human-friendly concepts with the corresponding fruit objects.

4. **Image Captioning** **April'22**
As part of the "CS5007: Deep Learning" coursework at IIT Palakkad, I built a deep learning framework to automatically generate natural-language captions for images. A pretrained CNN was used to extract image embeddings, which were then fed into an LSTM to generate the corresponding captions. Model performance, evaluated using the BLEU score, showed results that were sufficiently good compared to benchmark models, and multiple CNN variants were experimented with to improve captioning accuracy.

5. **Automatic Summarization Using Semantic WordClouds** **May'18 - July'19**
As part of my MTech thesis, I developed a research project to visualize natural language text summaries using Word Clouds, a text-mining technique that extracts and displays a document's most prominent keywords. The work focused on semantic word clouds, which position words to reflect their relationships rather than placing them randomly, proposing a novel methodology that preserves local word neighborhoods from the source text. The approach involved preprocessing the source text and ranking words by occurrence, then generating a semantic adjacency matrix between top-ranked words using word co-occurrence. K-Medoids clustering was applied to this similarity matrix to group related words. Finally, words were visualized using a geometric layout—placing the top keyword at the canvas centre and positioning subsequent words by approximating minimum similarity distance to already-placed words, while retaining cluster-assigned colours. We evaluated our method using the metric "Realized Adjacencies" and we obtained satisfactory results for our method as compared to random wordclouds.

PH.D COURSE WORK OVERVIEW

COURSE	GRADE
CS5512 : Machine Learning	S
CS5011 : Optimization	S
CS5007 : Deep Learning	S
CS6006 : Responsible AI	S
MA5007 : Probability & Statistics	A
GN6001 : Research Methodology	P (Pass/Fail - 0 credits)
CGPA	**9.75/10**
Total Credits Earned	**16**

RESEARCH POSTERS/TALKS

• Presented posters of our published work "HEIGHTS" at the following venues at IIT Palakkad
- IAC 5.0 (Industry Academia Conclave, 2023),
- Annual Research Scholars' Day 2024,
- Annual Data Science Research Symposium 2026,
- IAC 8.0 (Industry Academia Conclave, 2026)

• Awarded with the "**Best Research Talk**" from the Department of Data Science in the Annual Research Scholars' Day 2025 at IIT Palakkad.

• Presented a poster of our ongoing work "FILTER" at the following venues at IIT Palakkad
- Annual Research Scholars' Day 2026

REVIEWER ROLES

ICML 2024; ComSys 2025, 2026; ICDM 2026; ICDM 2026

MASSIVE OPEN ONLINE COURSES (MOOC)

1. Machine Learning - **Stanford University**
2. Mathematics for Machine Learning: Linear Algebra - **Imperial College London**
3. Introduction to Data Science in Python - **University Of Michigan.**
4. Data Collection And Processing with Python - **University Of Michigan**
5. Python Functions, Files And Dictionaries - **University Of Michigan**
6. Python Basics - **University Of Michigan.**

WORKSHOPS ATTENDED

1. Data Science: Probabilistic and Optimization Methods	**ICTS (TIFR) Bangalore**
2. Matlab for Machine Learning	**Mathworks & IIT Palakkad**
3. Applications of AI for Predictive Maintenance	**NVIDIA Deep Learning Institute**
4. Accelerating Data Engineering Pipelines	**NVIDIA Deep Learning Institute**
5. Fundamentals of Deep Learning	**NVIDIA Deep Learning Institute**
6. Technical Writing Workshop	**IIT Palakkad & IIT Madras**
7. Sun Certified Programmer for Java 6	**Oracle India Private Limited**

WORKSHOPS CONDUCTED

1. Petrichor TechFest 2022, IIT Palakkad
- Conducted one session of a hands-on workshop for external participants.
- Provided the basic concepts of machine learning, such as Linear regression, Ridge regression, Classification, Support Vector Machines, and Clustering.

2. ScienceQuest (AI Workshop for school students), IIT Palakkad, 2023 & 2024
- Conducted workshop for school students on the basics of Machine Learning & Deep Learning

- Engaged students to develop, train and test a basic image classification model using *Google Teachable Machine*.

3. Time Series Workshop, IIT Palakkad, 2024

- Conducted a day-long workshop for external and internal participants.
- Provided hands-on sessions on various time series analysis techniques, such as forecasting, anomaly detection, statistical methods such as ARIMA, SARIMA, and sequential deep models, such as RNN, LSTM, and GRU.

4. AI & Data Science Session, Annual Open House Event, IIT Palakkad, 2025

- Conducted a session for external and internal participants.
- Provided a demonstration of various AI tools and techniques for common users and school students.

5. Data Analytics Certification Course, IIT Palakkad, 2026

- Conducted a session for external industry participants.
- Provided a demonstration of various techniques on Clustering & Assosciation Rule Mining

RESEARCH GROUP ACTIVITIES

Research Group: MLG IIT Palakkad

1. Florence Nightingale Datascience Talk Series

- Conducted monthly talks on various emerging AI topics by experts in academia and industry

2. Journal Reading Club Presentations

- Presented various breakthrough papers and topics relevant to the domain of AI and Data Science, such as Transformer, TimeSeries Transformer, TranAD (Transformer for Anomaly Detection In Time Series Data), Normalizing Flow, Reinforcement Learning, Q-Learning, Deep Q-Learning, Grover's Algorithm in Quantum Computing, KAN (Kolmogorov-Arnold Networks), KAN-AD (KAN For Time Series Anomaly Detection), ProtoAD (Prototypes for Explainable Time Series Anomaly Detection)

3. Website Maintenance

- Maintained and updated the website for the research group to have wider outreach

4. Server Management

- Handled the GPU servers specific to the research group and managed various of its operations for the members, such as user creation, user access management, and environment setup

TALKS ATTENDED

1. **Fundamental Limits of Learning Non-Hallucinating Generative Models**,
Prof. Ananth Grama, Purdue University.
2. **New Frontiers in AI and Biomedicine**,
Prof. Shankar Subramaniam, UC San Diego.
3. **LLMs to improve ads quality and relevance in Microsoft ads**,
Dr. Samarth Bharadwaj, Microsoft.
4. **Temporal Graph Pipelines: From Algorithmics to Predictive Models**,
Prof. Srikanta Bedathur, IIT Delhi.
5. **Knowledge-Driven AI: Ontologies, Knowledge Graphs, and Explainable Systems**,
Dr. Raghava Mutharaju, IIT Palakkad.
6. **AI for Data Systems, Data Systems for AI: Toward Semantic Infrastructure for the Next Wave of Intelligence**,
Dr. Anand Eswaran, IBM Research.
7. **Efficient, Robust, and Equitable: A Decade of Deep Learning**,
Prof. R. Venkatesh Babu, IISc Bangalore.
8. **AI-in-the-loop for healthcare**,
Prof. Sriraam Natarajan, University of Texas.
9. **Different Approaches to Natural Language Processing**,
Mr. Siddhartha Mukherjee, Director, Samsung R&D Institute,.
10. **Practical applications of Large Language Models**,
Dr. Ashish Tendulkar, Google.
11. **An Information-theoretic Lens for Spectral Analysis**,
Dr. Aditi Kathpalia, IIT Madras.

12. **The Promise and Practicalities of AI in Engineered Systems**,
Ms. Nikita Pinto, Mathworks.
13. **The Power of Pretraining**,
Prof. Niloy Ganguly, IIT Kharaghpur.
14. **Vignettes of AI research: Foundations of Unsupervised Learning and Social Robotics**,
Prof. Chiranjib Bhattacharyya, IISC Bangalore.
15. **When Algorithms Meet Self-Interest: Foundations of Multi-Agent AI**,
Dr. Swaprava Nath, IIT Bombay.
16. **Data-driven decision frameworks for COVID-19 response**,
Prof. Rajesh Sundaresan, IISC Bangalore.
17. **Explainability is All You Need: The Saga of AI and Law**,
Dr. Kripabandhu Ghosh, IISER Kolkata.
18. **Question Answering using Knowledge Graphs**,
Dr. Soumajit Pramanik, IIT Bhilai.
19. **The Promise and Perils of AI: Social & Economic**,
Prof. Anjula Gurtoo, IISC Bangalore.

ACADEMIC ACHIEVEMENTS

1. UGC-NET JRF Award with 100 percentile score, December 2019.
2. UGC-NET Assistant Professor Qualification, December 2019
3. Academic Excellence Award, College Of Engineering Trivandrum, 2019.
4. GATE 2017: All India Rank - 826; Score : 684.
5. AIEEE 2004: All India Rank - 2920; Kerala Rank: 97.
6. Kerala Engineering Entrance, 2004 : Rank 151.

SKILLS

Languages: Python, MATLAB, Java, HTML, C, C++
Frameworks: PyTorch, TensorFlow, Numpy, Pandas, Scikit-learn, Matplotlib, Seaborn, Optuna, Ray Tuning
Tools: Spyder, Jupyter Notebook, Visual Studio, NetBeans, Google Colab
Environment : Conda

TEACHING ASSISTANCE (IIT PALAKKAD)

DS2010: Optimization	Aug - Dec '22, Aug - Dec '23
DS5608: Time Series Analysis	Jan - May '24
DS5612: Data Mining	Aug - Dec '24
DS3030: Data Analytics	Aug - Dec '25
DS3060: AI Ethics	Jan - May '25, Jan - May '26

Audience : Master & Undergraduate students

Tasks Done :

1. Designed tutorials, homework sets, and programming assignments with novel problems to strengthen students' grasp of core concepts, alongside illustration-based quizzes that promoted creative thinking over rote learning. Also responsible for setting midterm questions, assignments, and their evaluation.

2. Developed Data Analysis lab assignments exposing students to diverse dataset types—numerical, tabular, language, and vision; tools such as WEKA. Assignments covered techniques including visualization, dimensionality reduction, clustering, and statistical analysis.

3. Developed the AI Ethics Lab project to analyze societal biases in the COMPAS dataset using fairness metrics such as Treatment Equality, Demographic Parity, and Equal Opportunity. Applied and evaluated various pre-, in-, and post-processing bias mitigation techniques to assess their effectiveness in reducing identified biases.

TEACHING EXPERIENCE

Institute: Mar Baselios College Of Engineering And Technology, Trivandrum, Kerala
Designation : Assistant Professor Jan'20 - July'20
Courses Taught : Software Engineering, Advanced Computer Architecture
1. Theoretical concepts were taught with visual aids and problems to ensure student participation and discussion
2. Provided group assignments on various software engineering tools such as BitMatrix24, OrangeScrum, and GitHub to enhance teamwork and innovation

SOFTWARE INDUSTRY EXPERIENCE

TECHNOLOGY FIRM : Oracle India Private Limited

ROLES :

Senior Applications Engineer June'12 - Sept'15
Technopark, Trivandrum, Kerala, India.

Applications Engineer June'09 - May'12
HITEC City, Hyderabad, Telangana, India.

Associate Applications Engineer May'08 - May'09
HITEC City, Hyderabad, Telangana, India.

PRODUCTS :

- **Application Management Pack** May'08 - May'12
  - Plugin between two cloud services "Oracle Enterprise Manager" and "Oracle E-Business Suite"
  - Developed efficient methods for seamless integration, irrespective of the service's version upgrades
- **CRM On Demand** June'12 - Sept'15
  - Cloud application which had to be maintained for providing the services of "Oracle Siebel CRM"

PERSONAL PROFILE

Date of Birth	10 April 1986
Gender	Female
Nationality	Indian
Address Of Correspondence	Aswathi(H) Anickadu P.O., Kottayam, Kerala, India. 686503.
Languages known	English, Malayalam, Hindi, Tamil, Telugu
Phone	+91 9446382600
Official Mail	112114001@smail.iitpkd.ac.in
Personal Mail	vinithamr.nitc@gmail.com

DECLARATION

I hereby declare that all the information contained in this resume is in accordance with facts or truths to my knowledge. I take full responsibility for the correctness of the said information.

PLACE : Palakkad, Kerala, India
DATE : October 12, 2026 Vinitha M Rajan